

Instructions

Git & GitHub: First Contact

Welcome to your first experience with **Git** and **GitHub**!

GitHub is used by software developers, researchers, and data scientists to collaborate on projects without overwriting each other's work. Today you'll learn the same workflow used on real collaborative projects.

This activity takes about **20–30 minutes** and can be completed entirely in your web browser.

By the end of this activity, you will have:

- Created a GitHub account (if needed)
- Made your first edit to a shared repository
- Submitted your changes as a **pull request**
- Seen how pull requests are merged

Before you begin

You'll see some new terminology today, including **repository**, **fork**, **commit**, **pull request**, and **merge**. Don't worry about memorizing these terms yet—the goal is to experience the workflow first.

You will...	GitHub term
Make your own copy of a repository	Fork
Save your changes	Commit
Ask to include your changes	Pull request
Accept the changes	Merge

Step 1: Create a GitHub account

If you do not already have a GitHub account:

1. Go to <https://github.com>.
2. Click **Sign up** and follow the prompts.
3. Verify your email address if prompted.

Tip: Choose a username that you would be happy to use professionally. Many students include their GitHub profile in job or graduate school applications.

Step 2: Open the class repository

Your instructor will provide a link to the course repository.

Open the repository, then click on the shared file (for example, `students.qmd`).

Step 3: Fork the repository

You do not have permission to edit the course repository directly. Instead, GitHub will create your own personal copy, called a **fork**.

1. Click the **pencil** icon.
2. When prompted, click **Fork this repository**.
3. GitHub will create your copy and open the file for editing.

From this point on, you're editing your own copy of the repository.

Step 4: Edit the file

Add your own row to the table using the same format as the existing entries.

For example:

```
| Jane Doe | janedoe123 | Has climbed 3 mountains |
```

Include:

- Your name
- Your GitHub username
- A fun fact about yourself

Only add your own row—do not edit anyone else's.

Step 5: Commit your changes

Scroll to the bottom of the page.

Under **Commit changes**, enter a short description of your edit (for example, **Add Jane Doe to student list**) and click **Propose changes**.

A **commit** is simply a saved version of your work together with a message describing what changed.

Step 6: Create a pull request

GitHub will now show a page to create a **pull request**.

A pull request asks the repository owner to review and merge your changes into the original repository.

Click **Create pull request**.

That's it! Your changes have been submitted.

Step 7: Watch the merges

Your instructor will demonstrate how pull requests are reviewed and merged into the shared repository, including what happens when multiple people edit the same line.

What you learned

Today you completed the basic GitHub workflow:

Fork → Edit → Commit → Pull Request → Merge

This is the same workflow used by software developers, researchers, and data scientists to collaborate on code, documents, and data.
